

Numbers: Aptitude Preparation Notes

Numbers are the fundamental units of mathematics. This document covers numbers, their types, important facts, divisibility rules, remainder facts, division theorems, and cyclicity for aptitude preparation.

Types of Numbers

- **Integers:** All numbers whose fractional part is 0 (zero).
 - *Examples:* -3, -2, 1, 0, 10, 100.
 - **Natural Numbers:** Counting numbers. All integers greater than 0.
 - *Examples:* 1, 2, 3, 4, 5, 6...
 - **Whole Numbers:** All natural numbers and 0 (zero).
 - **Prime Numbers:** All numbers having only two distinct factors: the number itself and 1.
 - *Examples:* 2, 3, 53, 67, 191.
 - **Composite Number:** All numbers greater than 1 that are **NOT** prime.
 - *Examples:* 4, 60, 91, 100.
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Important Points on Prime Numbers

- 0 and 1 are neither prime nor composite.
- 2 is the only even prime number.
- There are 25 prime numbers less than 100: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Prime Number Check Rule

To check if a number 'p' is prime:

1. Find a number 'n' such that 'n' is the smallest natural number that satisfies $n^2 \geq p$.
2. Check if 'p' is divisible by any of the prime numbers less than or equal to 'n'.
3. If 'p' is **NOT** divisible by any such prime numbers, 'p' is a prime number. Otherwise, 'p' is not a prime number.

Co-primes

Two numbers 'a' and 'b' are called co-prime if their highest common factor (HCF) is 1.

Count of Divisors

The count of divisors of a number 'n' can be found using prime factorization.

If the prime factorization of a number n is $p_1^{e_1} \times p_2^{e_2} \times p_3^{e_3} \times \dots \times p_k^{e_k}$, then the number of divisors of n is: $(e_1 + 1) \times (e_2 + 1) \times (e_3 + 1) \times \dots \times (e_k + 1)$

- *Example:* \$200\$ can be written as $2^3 \times 5^2$. The number of divisors of 200 is $(3 + 1) \times (2 + 1) = 12$.
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Divisibility Rules

Here's a table summarizing the divisibility rule of common numbers:

Divisibility by Number	Rule	Example
Divisibility By 2	The last digit should be even.	124 last digit is 4 (even), so it is divisible by 2.
Divisibility By 3	The sum of its digits should be divisible by 3.	12321 is divisible by 3 because $1 + 2 + 3 + 2 + 1 = 9$ and 9 is divisible by 3.
Divisibility By 4	The last two digits should be divisible by 4.	1232 is divisible by 4 as the last two digits 32 are divisible by 4.
Divisibility By 5	The last digit is either 0 or 5.	85 last digit is 5, so 85 is divisible by 5.
Divisibility By 6	The number should be divisible by both 2 and 3.	114 is divisible by 6 as it is divisible by both 2 (last digit 4 is even) and 3 ($1 + 1 + 4 = 6$, which is divisible by 3).
Divisibility By 7	Remove the last digit from the number. Double the last digit, subtract it from the rest of the number, and if the result is divisible by 7, the number is divisible by 7.	196 is divisible by 7: Remove last digit, get 19. Double the last digit: $6 \times 2 = 12$. Subtract: $19 - 12 = 7$. Since 7 is divisible by 7, 196 is divisible by 7.
Divisibility By 8	The last three digits should be divisible by 8.	1232 is divisible by 8 as the last three digits 232 is divisible by 8 ($232 \div 8 = 29$).
Divisibility By 9	The sum of its digits is divisible by 9.	12321 is divisible by 9 because $1 + 2 + 3 + 2 + 1 = 9$ and 9 is divisible by 9.
Divisibility By 11	The difference between the sum of numbers at even positions and odd positions is either 0 or a multiple of 11.	121 is divisible by 11: Sum of odd position digits $1 + 1 = 2$. Sum of even position digits 2 . Difference $2 - 2 = 0$, which is divisible by 11.

Other Useful Divisibility Rules

- If a three-digit number is repeated twice to form a six-digit number (e.g., 123123), the result will be divisible by 7, 11, and 13. Dividing the six-digit number by $7 \times 11 \times 13 = 1001$ will give the original three-digit number.
- A number of the form 2^N has exactly $N+1$ divisors.
 - *Example:* $4 = 2^2$ has $2+1=3$ divisors: 1, 2, and 4.
- **Rule for Co-primes:** If 'p' and 'q' are co-primes and a number 'n' is divisible by both 'p' and 'q', then 'n' will be divisible by $p \times q$.
 - *Example:* 48 is divisible by both 3 and 8 (co-primes), and also by $3 \times 8 = 24$.

- **Rule for Non-Co-primes:** If 'p' and 'q' are **NOT** co-prime, the fact that 'n' is divisible by both 'p' and 'q' does **NOT** necessarily mean 'n' is divisible by $p \times q$.
 - *Example:* 144 is divisible by both 8 and 12 (not co-prime), but it is not divisible by $8 \times 12 = 96$.

Remainder Facts

The table summarizes rules to quickly determine the remainder without performing full division.

Remainder when Dividing by	Rule	Example
2	If the last digit is odd, the remainder is 1. If it's even, the remainder is 0.	37 $\rightarrow$ last digit is 7 (odd), so remainder = 1.
3	Sum the digits of the number. Find the remainder of this sum when divided by 3.	23 $\rightarrow$ Sum of digits $2 + 3 = 5$. $5 \div 3$ gives remainder 2. So, remainder = 2.
4	Take the last two digits of the number, divide them by 4, and the remainder is the remainder of the original number.	123 $\rightarrow$ Last two digits 23. $23 \div 4$ gives remainder 3. So, remainder = 3.
5	Divide the last digit of the number by 5 to get the remainder.	32 $\rightarrow$ Last digit is 2. $2 \div 5$ gives remainder 2. So, remainder = 2.
6	Apply both the remainder rules for 2 and 3. If both are 0, the remainder is 0. Otherwise, you have to calculate accordingly.	35 $\rightarrow$ Remainder when divided by 2 is 1. Remainder when divided by 3 is 2. The remainder when divided by 6 is 5.
7	For smaller numbers: Break the number into powers of 10 and calculate modulo 7 for each. The remainders of $10^0, 10^1, 10^2, \dots$ when divided by 7 are 1, 3, 2, 6, 4, 5, 1, $\dots$	198 $\rightarrow (1 \times 10^2) + (9 \times 10^1) + (8 \times 10^0) \equiv (1 \times 2) + (9 \times 3) + (8 \times 1) \pmod{7} \equiv 2 + 27 + 8 \pmod{7} \equiv 2 + 6 + 1 \pmod{7} \equiv 9 \pmod{7}$. Remainder is 2.
8	Take the last three digits, divide by 8, and the remainder is the remainder of the original number.	1,476 $\rightarrow$ Last three digits 476. $476 \div 8$ gives remainder 4. So, remainder = 4.
9	Sum the digits of the number. The remainder when this sum is divided by 9 is the remainder of the original number.	1234 $\rightarrow$ Sum of digits $1 + 2 + 3 + 4 = 10$. $10 \div 9$ gives remainder 1. So, remainder = 1.

Division Theorem

The fundamental relationship between the dividend, divisor, quotient, and remainder is:

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

Divisibility Properties (Binomial Form)

- $x^n - a^n$ is divisible by $(x - a)$ for all values of n .
 - Example ($n=2$):* $x^2 - a^2 = (x - a)(x + a)$, which is divisible by $(x - a)$.
 - Example ($n=3$):* $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$, which is divisible by $(x - a)$.
 - $x^n - a^n$ is divisible by $(x + a)$ for all **even** values of n .
 - Example ($n=2$):* $x^2 - a^2 = (x - a)(x + a)$, which is divisible by $(x + a)$.
 - Example ($n=3$):* $x^3 - a^3$ is not divisible by $(x + a)$.
 - $x^n + a^n$ is divisible by $(x + a)$ for all **odd** values of n .
 - Example ($n=3$):* $x^3 + a^3 = (x + a)(x^2 - ax + a^2)$, which is divisible by $(x + a)$.
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Cyclicity of Numbers (Unit Digits)

The cyclicity of any number is mainly focused on the repetitive pattern of its unit digit when raised to any power.

Digits with Cyclicity of 1 (Unit digit repeats itself)

- Digits:** 0, 1, 5, and 6.
- When each of these digits is raised to any power, the unit digit of the final answer is the number itself.
 - Examples:*
 - $5^2 = 25$: Unit digit is 5.
 - $1^6 = 1$: Unit digit is 1.
 - $0^4 = 0$: Unit digit is 0.
 - $6^3 = 216$: Unit digit is 6.

Digits with Cyclicity of 2

- Digits:** 4 and 9.
- For 4:* Unit digits alternate between 4 and 6.
 - $4^1 = 4$
 - $4^2 = 16$ (Unit digit is 6)
 - $4^3 = 64$ (Unit digit is 4)
- For 9:* Unit digits alternate between 9 and 1.
 - $9^1 = 9$
 - $9^2 = 81$ (Unit digit is 1)
 - $9^3 = 729$ (Unit digit is 9)

Digits with Cyclicity of 4

- Digit 2:** The cycle of unit digits is 2, 4, 8, 6.
- Digit 3:** The cycle of unit digits is 3, 9, 7, 1.
- Digit 7:** The cycle of unit digits is 7, 9, 3, 1.
- Digit 8:** The cycle of unit digits is 8, 4, 2, 6.